

BOSCOBEL, WI, USAWMO: **726438**Lat: **43.156N**Lon: **90.678W**Elev: **205**StdP: **98.89**Time Zone: **-6.00 (NAC)**Period: **01-19**WBAN: **94994****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-20.0	-27.5	0.3	-22.1	-24.8	0.4	-19.5	8.3	-3.7	7.6	-4.1	1.2	220	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	33.1	23.9	31.7	22.9	30.2	22.1	25.3	31.3	24.4	29.8	23.4	28.4	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	18.9	28.4	22.6	17.7	27.5	21.8	16.9	26.6	78.5	31.1	74.6	29.9	70.7	28.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.8	6.9	5.7	-28.7	35.5	3.4	1.7	-31.1	36.7	-33.1	37.7	-35.1	38.7	-37.5	39.9
			-28.7	27.0	2.9	1.0	-30.7	27.8	-32.4	28.4	-34.0	29.0	-36.1	29.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-6.7	-5.2	2.1	9.2	15.3	20.7	22.9	21.7	17.7	10.2	3.4	-3.4
	DBStd	11.62	7.09	6.70	6.67	5.32	4.83	3.43	3.16	3.03	4.40	5.09	5.44	6.20
	HDD10.0	1972	518	424	257	77	9	0	0	0	2	61	207	417
	HDD18.3	3802	776	658	503	278	119	16	3	6	64	256	448	675
	CDD10.0	1632	0	0	13	52	174	320	400	363	233	67	9	1
	CDD18.3	421	0	0	1	3	26	86	144	111	44	4	0	0
	CDH23.3	4501	0	0	12	68	349	929	1528	1063	482	68	1	0
	CDH26.7	1588	0	0	1	17	109	324	613	361	149	14	0	0
Wind	WSAvg	2.2	2.4	2.5	2.5	2.9	2.4	1.9	1.7	1.5	1.8	2.3	2.4	2.3
Precipitation	PrecAvg	887	25	28	50	96	108	131	105	111	82	58	50	40
	PrecMax	1240	52	54	95	170	275	230	224	427	204	144	116	80
	PrecMin	622	9	10	21	23	43	30	39	24	14	16	5	8
	PrecStd	156	12	14	25	41	56	50	48	91	49	35	30	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	14.6	25.3	29.0	32.6	34.3	36.6	33.9	32.9	28.7	21.7	14.9
		MCWB	5.4	10.6	17.2	17.4	22.0	24.1	25.6	23.8	23.4	19.5	14.8	12.4
	2%	DB	7.2	9.1	19.4	25.1	29.5	32.1	33.5	32.0	30.5	25.2	17.7	9.4
		MCWB	4.2	5.6	13.3	15.4	19.7	23.0	24.6	23.5	22.4	17.4	12.1	6.8
	5%	DB	4.5	6.3	16.1	22.3	27.3	30.1	31.8	30.2	28.1	22.4	15.0	7.1
		MCWB	2.3	3.5	11.3	13.6	18.4	21.6	23.4	22.4	20.5	16.5	10.7	4.6
	10%	DB	2.7	3.8	12.2	19.5	24.8	28.4	30.0	28.7	26.3	19.5	12.3	4.8
		MCWB	1.2	1.4	7.6	11.5	17.1	20.8	22.6	21.6	19.3	14.8	8.5	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	9.5	17.8	18.8	23.3	25.8	27.3	25.7	24.9	21.9	15.2	12.4
		MCDB	9.3	14.0	24.6	26.5	30.8	31.9	33.0	31.5	30.9	26.0	19.8	14.2
	2%	WB	4.2	6.0	15.0	16.5	21.4	24.4	25.7	24.7	23.2	19.5	13.3	7.1
		MCDB	6.9	8.5	18.2	23.2	27.1	30.4	31.6	29.8	28.4	22.8	17.0	8.5
	5%	WB	2.7	3.6	11.0	14.8	19.8	23.1	24.7	23.7	21.9	17.5	11.1	4.8
		MCDB	4.3	6.2	15.0	20.8	25.0	28.3	30.4	28.3	26.1	21.5	14.4	6.6
	10%	WB	1.1	1.7	7.9	12.7	18.3	21.8	23.7	22.7	20.6	15.0	8.7	2.9
		MCDB	2.7	3.5	12.0	18.6	23.3	26.5	28.6	27.0	24.7	19.0	11.8	4.7
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.8	11.4	12.9	13.3	12.6	12.4	13.2	12.3	10.4	9.0	
		MCDBR	11.4	13.0	15.8	17.9	16.3	14.9	14.1	14.0	15.3	16.5	15.0	11.8
		MCWBR	8.8	9.7	9.6	9.5	7.6	7.0	6.1	6.2	7.5	9.2	10.0	9.4
	5% WB	MCDBR	10.9	12.7	14.1	15.0	13.3	12.6	12.3	11.6	12.4	13.6	13.6	10.5
		MCWBR	8.7	9.8	9.4	8.9	6.9	6.6	6.0	5.8	6.7	8.9	9.9	9.1
Clear-Sky Solar Irradiance	taub	0.275	0.303	0.329	0.374	0.400	0.423	0.428	0.418	0.380	0.338	0.296	0.277	
	taud	2.383	2.313	2.385	2.304	2.279	2.252	2.242	2.272	2.358	2.472	2.515	2.436	
	Ebn at Noon	867	893	913	892	873	851	842	838	846	846	841	832	
	Edh at Noon	77	99	105	124	131	135	135	127	108	85	69	68	
All-Sky Solar Radiation	RadAvg	1.78	2.83	3.83	4.55	5.40	6.00	6.10	5.27	4.32	2.78	1.84	1.36	
	RadStd	0.21	0.27	0.44	0.39	0.40	0.46	0.34	0.36	0.35	0.35	0.15	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page